

- NOTE:**
1. All questions are compulsory.
 2. Mark answers number clearly.
 3. Do not write anything on the question paper except Name and Roll Number.

S. No.	Question	Marks
1.	(a) Let A be a 3×3 matrix with eigenvalues 1, 2, 3 and corresponding eigenvectors $x_1 = (2, -1, 0)$, $x_2 = (-1, 2, -1)$, $x_3 = (0, -1, 2)$. (i) Find the matrix A. (ii) Let $B = (A - I)(A - 2I)(A - 3I)$. Determine the eigenvectors and eigenvalues of B.	3+3
	(b) Let $\mathcal{P}_2$ be the vector space of all real polynomials of degree at most 2. Define a map $T : \mathcal{P}_2 \rightarrow \mathbb{R}^3, \quad T(p) = (p(0), p'(1), p''(2)).$ (i) Find the matrix representation of T with respect to the standard basis $\{1, x, x^2\}$ of $\mathcal{P}_2$ and the standard basis of $\mathbb{R}^3$. (ii) Find the basis of null-space (kernel) and image (range) space of T. (iii) Define a map $\langle \cdot, \cdot \rangle : \mathcal{P}_2 \times \mathcal{P}_2 \rightarrow \mathbb{R}$ by $\langle p, q \rangle = T(p) \cdot T(q),$ where “$\cdot$” in the last line above is the usual dot product. Determine whether this defines an inner product on $\mathcal{P}_2$.	2+2+2

S. No.	Question	Marks
2.	(a)(i) Show that the divergence of the curl of any vector field is identically zero. Similarly, show that the curl of the gradient of any scalar field is identically zero. Use the standard definitions of these operations and appropriate properties of partial derivatives in your argument. (ii) Let the set $\{a, b, c\}$ be a basis of $\mathbb{R}^3(\mathbb{R})$ and $d = \alpha a + \beta b + \gamma c$, where α, β, γ are real numbers. Prove that the set $\{a - d, b - d, c - d\}$ is a basis of $\mathbb{R}^3(\mathbb{R})$ if and only if $\alpha + \beta + \gamma \neq 1$.	3+3
	(b) Verify Green's theorem in the plane for the function $\mathbf{F}(x, y) = (2xy - x^2, x + y^2)$ and the region enclosed by the parabola $y = x^2$ and $y^2 = x$.	6
3.	(a)(i) Find the region of convergence of the series $\sum_{n=-\infty}^{\infty} 3^{- n } z^{2n}$. (ii) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in a Laurent series valid for: $1 < z < 3$ and $0 < z+1 < 2$.	3+3
	(b) Let $f(z) = \begin{cases} \frac{\operatorname{Im}(z^2)}{ z ^2}, & z \neq 0, \\ 0, & z = 0. \end{cases}$ Check whether $f(z)$ satisfies the Cauchy-Riemann equations at $z = 0$ or not. Also, check the analyticity of $f(z)$ at $z = 0$.	6

S. No.	Question	Marks
4.	(a) For each of the following functions, find all singularities, determine their nature, and compute the residue: (i) $f(z) = (1 - z^3)e^{1/z}$ (ii) $f(z) = \frac{\sin z}{z^{2010}}$	3+3
	(b) State Cauchy's Residue Theorem. Evaluate the integral $\oint_{ z =2} \frac{z - 3 \cos z}{(z - \frac{\pi}{2})^2} dz.$	2+4
5.	(a) Evaluate the integral by using the method of residues $I = \int_{-\infty}^{\infty} \frac{dx}{x^4 + 1}$	6
	(b) A physicist studies a vector field $\mathbf{F}(x, y, z)$. From experiments, it is known that $\mathbf{F}$ is of the form $\mathbf{F}(x, y, z) = xz \hat{\mathbf{i}} + (axe^y z + byz) \hat{\mathbf{j}} + (y^2 - xe^y z^2) \hat{\mathbf{k}},$ where a and b are real constants. It is further known that $\mathbf{F} = \nabla \times \mathbf{G}$ for some differentiable vector field $\mathbf{G}$. (i) Use a known property of divergence for vector fields of the form $\nabla \times \mathbf{G}$ to determine a and b. (ii) Evaluate the surface integral $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} dS,$ where S is the part of the ellipsoid $x^2 + y^2 + \frac{1}{4}z^2 = 1$, $z \geq 0$, oriented so that the normal vector has a positive z-component. (iii) Let S_1 be the closed surface obtained by adjoining to S the disk in the plane $z = 0$. Find the total outward flux and comment on your result.	2+2+2

ALL THE BEST